Stochastic Finance and Risk Modelling

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2 - From probability to stochastic process : continuation

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Outline

Stochastic process

Filtration

Martingale

Program

Stochastic process

Filtration

Martingale

Stochastic process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Definition (stochastic process)

A discrete-time stochastic process is a family of r.v.s $X = (X_t)_{t \in I}$ defined on $(\Omega, \mathcal{F})$ and taking values in $\mathbb{R}^d$, where $I = \{0, 1, \ldots, T\}$ is the index set with $T \in \mathbb{N}$ or $T = \infty$.

- Throughout this course we consider the discrete-time setting $I = \{0, 1, \ldots, T\}$.
- For the continuous-time case $I = [0, T]$ or $I = \mathbb{R}$, see e.g. "Calcul Stochastique 3A".

Head or tail

- Game : Throw successively a balanced coin T times, with H for head and T for tail. Denote $E := \{head, tail\}$.
- Probability space : $\Omega := E^T = \{\omega = (\omega_1, \ldots, \omega_T) : \ \omega_i \in E\}$, $\mathcal{F} := 2^\Omega = \{A : A \subset \Omega\}$ and $\mathbb{P}(\omega) := 1/|\Omega| = 1/2^T$.
- Let $X_1, \ldots, X_T$ be r.v.s defined by $X_i(\omega) = 1$ if $\omega_i = head$ and $X_i(\omega) = 0$ if $\omega_i = tail$. Then $(X_i)_{1 \leq i \leq T}$ is a stochastic process taking values in $\{0, 1\}$.
- Let $S_t := \sum_{i=1}^{t} X_i$. Then $(S_n)_{1 \leq t \leq T}$ is a stochastic process taking values in $\{0, \ldots, T\}$.

Trajectories

$(X_t)_{t\in I}$ is a stochastic process defined on $(\Omega, \mathcal{F}, \mathbb{P})$. Set $\Sigma := \mathbb{R}^d$ and $\mathcal{G} := \mathcal{B}(\mathbb{R}^d)$

- View 1 :

$$\begin{aligned} X : \left(I \times \Omega, 2^I \otimes \mathcal{F}\right) &\to (\Sigma, \mathcal{G}) \\ (t, \omega) &\mapsto X_t(\omega), \end{aligned}$$

- View 2 : let Σ^I be the set of function from I to Σ

$$\begin{aligned} X : (\Omega, \mathcal{F}) &\to (\Sigma^I, \otimes_I \mathcal{G}) \\ \omega &\mapsto (X_t(\omega))_{t\in I}. \end{aligned}$$

Definition (trajectory)

For each $\omega \in \Omega$, we call $(X_t(\omega))_{t\in I}$ a trajectory or path of X. Each path stands for a realization of the stochastic process.

A process is seen as a random variable taking values in Σ^I.

Program

Stochastic process

Filtration

Martingale

Filtration

Definition (Filtration)

A filtration is an increasing sequence of sub-algebras $(\mathcal{F}_t)_{t\in I}$ of $\mathcal{F}$, i.e.

$$\mathcal{F}_t \subseteq \mathcal{F}_{t+1} \subseteq \mathcal{F}, \quad \forall 0 \leq t \leq T-1.$$

- Recall that $E = \{head, tail\}$, $\Omega = E^T$ and $\mathcal{F} = 2^\Omega$.
- Set $\mathcal{F}_0 := \{\emptyset, \Omega\}$ and $\mathcal{F}_t := \sigma(X_1, \ldots, X_t)$ for $1 \leq t \leq T$.
- $(\mathcal{F}_t)_{0\leq t\leq T}$ is a filtration for $(\Omega, \mathcal{F}, \mathbb{P})$.
- $\mathcal{F}_t = \sigma(S_1, \ldots, S_t)$ for $1 \leq t \leq T$.
- $\mathcal{F}_t$ stands for the accumulated information up to time t.

Adapted/Predictable process

Definition (adapted/predictable process)

X is called adapted (resp. predictable) w.r.t. to filtration $(\mathcal{F}_t)_{t\in I}$ if X_t (resp. X_{t+1}) is $\mathcal{F}_t-$measurable for all $t \in I$.

- Adapted processes are used to model the events to come out, like prices.
- Predictable processes are used to model a decision made for the near future $t+1$, provided the information up to time t, like investing strategies.

Proposition

If X is adapted, then X_s is $\mathcal{F}_t-$measurable for all $s \leq t$.

Generated filtration

Definition

We call $\mathcal{F}^X$ the filtration generated by X which is defined as

$$\mathcal{F}_t^X := \sigma(X_0, \ldots, X_t).$$

Proposition

X is adapted w.r.t. its generated filtration.

- $\mathcal{F}^X$ is the smallest filtration that makes X adapted.

Stopping time

Definition (stopping time)

A random variable τ taking values in $\mathbb{N} \cup \{\infty\}$ is called a stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \in \mathbb{N}$.

Proposition

- *A random variable τ taking values in $\mathbb{N} \cup \{\infty\}$ is a stopping time if and only if $\{\tau = t\} \in \mathcal{F}_t$ for all $t \in \mathbb{N}$.*
- *Let $A \in \mathcal{B}(\mathbb{R}^d)$ and X be adapted. Then the first hitting time $\tau := \inf\{t \in I : X_t \in A\}$ is a stopping time, where $\inf \emptyset := \infty$.*

Stopped process

Definition (stopped process)

Let X be an adapted process and τ be a stopping time. Then, $X^\tau := (X_{t\wedge\tau})_{t\in I}$ is called the stopped process by τ. Moreover, X^τ is adapted.

$$X_t^\tau(\omega) = X_{t\wedge\tau(\omega)}(\omega) = \begin{cases} X_t(\omega) \text{ on } \{t < \tau(\omega)\} \\ X_{\tau(\omega)}(\omega) \text{ on } \{t \geq \tau(\omega)\}. \end{cases}$$

Program

Stochastic process

Filtration

Martingale

Discrete-time martingale

Let $(X_t)_{t\in I}$ be adapted and integrable, i.e. X_t is $\mathcal{F}_t$-measurable and $\mathbb{E}[|X_t|] < \infty$ for all $t \in I$.

Definition (martingale)

X is called

(i) sub-martingale if $\mathbb{E}[X_t|\mathcal{F}_s] \geq X_s$ a.s. for all $t, s \in I$ such that $t \geq s$;

(ii) super-martingale if $\mathbb{E}[X_t|\mathcal{F}_s] \leq X_s$ a.s. for all $t, s \in I$ such that $t \geq s$;

(iii) martingale if $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$ a.s. for all $t, s \in I$ such that $t \geq s$.

It is enough to show the claims with $t = s + 1$ for all $s \in I$ (such that $s + 1 \in I$). In the case of a martingale and if $I = \{0, \ldots, T\}$ it is enough to prove that $\mathbb{E}[X_T|\mathcal{F}_s] = X_s$ for all $s \in I$.

Proposition

If X is martingale, $\mathbb{E}[X_t] = \mathbb{E}[X_0]$ for all $t \in I$. An adapted and integrable process X is martingale iff $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$ for all bounded stopping times τ.

Doob's martingale inequality

Theorem

Let X be sub-martingale. Then for any $x > 0$, one has

$$\mathbb{P}(\max_{1\leq i\leq n} X_i > x) \leq \frac{\mathbb{E}[X_n^+]}{x}.$$

If X is L^p-integrable martingale with $p > 1$, then

$$\mathbb{P}(\max_{1\leq i\leq n} X_i > x) \leq \frac{\mathbb{E}[|X_n|^p]}{|x|^p} \qquad \mathbb{E}[(\max_{1\leq i\leq n} |X_i|)^p] \leq \frac{p^p}{(p-1)^p}\mathbb{E}[|X_n|^p].$$

Doob's martingale convergence theorem

Theorem

Let $T = \infty$ and X be a super-martingale s.t.

$$\sup_{t \in I} \mathbb{E}[X_t^-] < \infty.$$

Then there exists $X_\infty \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ s.t.

$$\lim_{t \to \infty} X_t = X_\infty \text{ a.s.}$$